

PHYS 427: Lecture # 8

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Lecture Span

1. Heat and Work

Recap

Micro canonical ensemble: fixed energy, number of particles. Canonical ensemble has a reservoir, combined energy and temperature are constant.

For MCE, $S(U, V)$ is maximized in equilibrium. For CE, $F(U, V)$ is minimized in equilibrium.

For MCE, $\Omega(U) = \sum_{\text{all states with energy } U} 1$.

For CE, $Z(T) = \sum e^{-\epsilon_i/k_B T}$.

For MCE, $S = k_B \ln \Omega$.

For CE, $F = -k_B T \ln Z$.

For MCE,

$$dS = (\partial_U S)_V dU + (\partial_V S) dV \quad (1)$$

$$= \frac{1}{T} dU + \frac{p}{T} dV \quad (2)$$

For CE,

$$dF = (\partial_T F)_T dT + (\partial_V F)_T dV \quad (3)$$

$$= -S dT - p dV \quad (4)$$

You can match definitions to then rewrite T in terms of S , p in terms of F , and etc.

Heat and Work

Thermodynamic identity:

$$dU = T dS - p dV \quad (5)$$

1. Work: can be electrical, or mechanical
2. Heat: cannot be completely converted into work because of the 2nd law of thermodynamics

The 2nd law of thermodynamics states $\Delta S \geq 0$, and this is the total entropy of a system!

Consider a reservoir of temperature T_H , then it takes heat Q_H , then outputs Q_C (heat that goes into the cold reservoir) and W . Then

$$Q_H = W + Q_C \quad (6)$$

Then

$$\Delta S_{tot} = \Delta S_{hot} + \Delta S_{cold} \geq 0 \quad (7)$$

$$= -\frac{Q_H}{T_H} + \frac{Q_C}{T_C} \geq 0 \quad (8)$$

The bottom reservoir then must be colder than the top reservoir. Define efficiency,

$$\eta = \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H} \quad (9)$$

The maximum possible efficiency for $\Delta S_{tot} = 0$, then

$$\frac{Q_C}{Q_H} = \frac{T_C}{T_H} \quad (10)$$

Then, the max efficiency is

$$\eta_{\max} = 1 - \frac{T_C}{T_H} \quad (11)$$

This is the Carnot efficiency. But you can have losses along the way, such as $\Delta S_{tot} > 0$, which can be caused by a heat leak from H to C.

Irreversible processes

Create $\Delta S_{tot} > 0$ that cannot be removed due to the 2nd law of thermodynamics.

Carnot Cycle

Idealized way to realize a heat engine at the maximal efficiency $\eta_{\max}$. Start at (T_H, S_1) , then heat up to (T_H, S_2) . Then go (T_H, S_2) to (T_C, S_2) . Then to (T_C, S_1) , then close the loop.

Here,

$$\oint dQ = (T_H - T_C)(S_2 - S_1) \quad (12)$$

The area of this rectangle is the work done by the engine. You can then calculate

$$\eta = \frac{W}{Q_H} = \frac{(T_H - T_C)(S_2 - S_1)}{T_H(S_2 - S_1)} = 1 - \frac{T_C}{T_H} \quad (13)$$

Note, reversible cooling and heating ($\Delta S = 0$) need to be very slow; otherwise, you gain entropy. But this is not practical.

Monatomic ideal gas as working substance

Sackur-Tetrode equation

$$S = Nk_B T \left(\ln \left(\frac{n_Q V}{N} \right) + \frac{5}{2} \right) \quad (14)$$

$PV = Nk_B T$. for isothermal, $PV = \text{const.}$

For isentropic,

$$S = Nk_B \left[\ln(T^{3/2}) + \ln(V) \right] \quad (15)$$

Then $T^{3/2}V$ is constant, which means that $p^{3/2}V^{5/2} = \text{const.}$ Then

$$pV^\gamma = \text{const.} \quad (16)$$

For monatomic ideal gas. For $\gamma = 5/3 = C_P/C_V$. For an ideal gas, you can plot it on a PV diagram.

First, isothermal expansion. Then, isentropic expansion. Then isothermal. Then isentropic. You can then calculate the work done by the system.

$$W = (T_H - T_C)(S_2 - S_1) \quad (17)$$

Irreversible expansion. Box with wall with door in the middle. Volume on left is V_i and on right is V_f . $Q = 0$. Also, $\Delta U = 0 \iff \Delta T = 0$. The entropy $\Delta S = Nk_B [\ln(V_f/V_i)] > 0$. For irreversible processes, $TdS > dQ$. For reversible, they are equal.